

## Hypothesis conclusion

The only way to make the conditional statement  $p \rightarrow q$  false is to \_\_\_\_\_

The **hypothesis** of  $p \rightarrow q$  is \_\_\_\_\_ The **antecedent** of  $p \rightarrow q$  is \_\_\_\_\_

The **conclusion** of  $p \rightarrow q$  is \_\_\_\_\_ The **consequent** of  $p \rightarrow q$  is \_\_\_\_\_

## Converse inverse contrapositive

The **converse** of  $p \rightarrow q$  is \_\_\_\_\_

The **inverse** of  $p \rightarrow q$  is \_\_\_\_\_

The **contrapositive** of  $p \rightarrow q$  is \_\_\_\_\_

# Logical operators english synonyms

## Common ways to express logical operators in English:

**Negation**  $\neg p$  can be said in English as

- Not  $p$ .
- It's not the case that  $p$ .
- $p$  is false.

**Conjunction**  $p \wedge q$  can be said in English as

- $p$  and  $q$ .
- Both  $p$  and  $q$  are true.
- $p$  but  $q$ .

**Exclusive or**  $p \oplus q$  can be said in English as

- $p$  or  $q$ , but not both.
- Exactly one of  $p$  and  $q$  is true.

**Disjunction**  $p \vee q$  can be said in English as

- $p$  or  $q$ , or both.
- $p$  or  $q$  (inclusive).
- At least one of  $p$  and  $q$  is true.

**Conditional**  $p \rightarrow q$  can be said in English as

- if  $p$ , then  $q$ .
- $p$  is sufficient for  $q$ .
- $q$  when  $p$ .
- $q$  whenever  $p$ .
- $p$  implies  $q$ .
- $q$  follows from  $p$ .
- $q$  is necessary for  $p$ .
- $p$  only if  $q$ .

**Biconditional**

- $p$  if and only if  $q$ .
- $p$  iff  $q$ .
- If  $p$  then  $q$ , and conversely.
- $p$  is necessary and sufficient for  $q$ .



# Definitions functions prereqs

| Term                 | Notation                                 | Example(s)                                                                                                                | We say in English ...                                                                                                                                                                                                                     |
|----------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sequence             | $x_1, \dots, x_n$                        |                                                                                                                           | A sequence $x_1$ to $x_n$                                                                                                                                                                                                                 |
| summation            | $\sum_{i=1}^n x_i$ or $\sum_{i=1}^n x_i$ |                                                                                                                           | The sum of the terms of the sequence $x_1$ to $x_n$                                                                                                                                                                                       |
| piecewise definition | rule                                     | $f(x) = \begin{cases} \text{rule 1 for } x & \text{when COND 1} \\ \text{rule 2 for } x & \text{when COND 2} \end{cases}$ | Define $f$ of $x$ to be the result of applying rule 1 to $x$ when condition COND 1 is true and the result of applying rule 2 to $x$ when condition COND 2 is true. This can be generalized to having more than two conditions (or cases). |
| function application |                                          | $f(7)$                                                                                                                    | $f$ of 7 <b>or</b> $f$ applied to 7 <b>or</b> the image of 7 under $f$                                                                                                                                                                    |
|                      |                                          | $f(z)$                                                                                                                    | $f$ of $z$ <b>or</b> $f$ applied to $z$ <b>or</b> the image of $z$ under $f$                                                                                                                                                              |
|                      |                                          | $f(g(z))$                                                                                                                 | $f$ of $g$ of $z$ <b>or</b> $f$ applied to the result of $g$ applied to $z$                                                                                                                                                               |
| absolute value       |                                          | $ -3 $                                                                                                                    | The absolute value of $-3$                                                                                                                                                                                                                |
| square root          |                                          | $\sqrt{9}$                                                                                                                | The non-negative square root of 9                                                                                                                                                                                                         |

**Pro-tip:** the meaning of two vertical lines  $| \quad |$  depends on the data-types of what's between the lines. For example, when placed around a number, the two vertical lines represent absolute value. We've seen a single vertical line  $|$  used as part of set builder definitions to represent "such that". Again, this is (one of the many reasons) why is it very important to declare the data-type of variables before we use them.